

FRACTIONS

Short Revision Notes — Remember Level (Recall Facts & Basic Knowledge)

1. Key Definitions

Term	Definition	Example
Fraction	A part of a whole	$\frac{1}{2}$, $\frac{3}{4}$, $\frac{2}{3}$
Numerator	The top number of a fraction (shows how many parts are taken)	In $\frac{3}{4}$, numerator = 3
Denominator	The bottom number of a fraction (shows total number of equal parts)	In $\frac{3}{4}$, denominator = 4
Proper Fraction	Numerator is smaller than denominator	$\frac{2}{3}$, $\frac{5}{8}$, $\frac{3}{4}$
Improper Fraction	Numerator is larger than or equal to denominator	$\frac{5}{3}$, $\frac{8}{5}$, $\frac{3}{3}$
Mixed Number	A whole number and a fraction combined	$1\frac{1}{2}$, $2\frac{3}{4}$, $3\frac{2}{3}$
Reciprocal	Flipping a fraction upside down (interchanging numerator and denominator)	Reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$

2. Types of Fractions

Type	Description	Examples
Proper Fraction	Numerator < Denominator (value < 1)	$\frac{1}{2}$, $\frac{1}{3}$, $\frac{3}{4}$, $\frac{5}{8}$
Improper Fraction	Numerator $\geq$ Denominator (value $\geq$ 1)	$\frac{5}{3}$, $\frac{7}{4}$, $\frac{11}{8}$, $\frac{3}{3}$
Mixed Number	Whole number + Proper fraction	$1\frac{1}{2}$, $2\frac{3}{4}$, $3\frac{5}{8}$, $5\frac{7}{8}$
Unit Fraction	Numerator = 1	$\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{4}$, $\frac{1}{5}$
Equivalent Fractions	Different fractions with the same value	$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8}$

3. Converting Mixed Numbers to Improper Fractions

Formula: Improper fraction = (Whole $\times$ Denominator + Numerator) / Denominator

Mixed Number	Calculation	Improper Fraction
$1\frac{1}{2}$	$(1 \times 2 + 1) / 2 = \frac{3}{2}$	$\frac{3}{2}$
$2\frac{3}{4}$	$(2 \times 4 + 3) / 4 = \frac{11}{4}$	$\frac{11}{4}$
$3\frac{2}{5}$	$(3 \times 5 + 2) / 5 = \frac{17}{5}$	$\frac{17}{5}$
$5\frac{7}{8}$	$(5 \times 8 + 7) / 8 = \frac{47}{8}$	$\frac{47}{8}$
$1\frac{3}{8}$	$(1 \times 8 + 3) / 8 = \frac{11}{8}$	$\frac{11}{8}$

Quick Trick: “Multiply, Add, Keep” — Multiply whole number by denominator · Add numerator · Keep the same denominator

4. Converting Improper Fractions to Mixed Numbers

Method: Divide numerator by denominator

Improper Fraction	Division	Mixed Number
$7/3$	$7 \div 3 = 2$ remainder 1	$2 \frac{1}{3}$
$19/4$	$19 \div 4 = 4$ remainder 3	$4 \frac{3}{4}$
$43/4$	$43 \div 4 = 10$ remainder 3	$10 \frac{3}{4}$
$36/7$	$36 \div 7 = 5$ remainder 1	$5 \frac{1}{7}$
$11/8$	$11 \div 8 = 1$ remainder 3	$1 \frac{3}{8}$

Quick Trick: “Divide, Remainder, Keep” — Divide numerator by denominator · The quotient is the whole number · The remainder is the new numerator · Keep the same denominator

5. Equivalent Fractions

Method: Multiply or divide both numerator and denominator by the SAME number

Original	Multiply by 2	Multiply by 3	Multiply by 4
$1/2$	$2/4$	$3/6$	$4/8$
$1/3$	$2/6$	$3/9$	$4/12$
$2/3$	$4/6$	$6/9$	$8/12$
$3/4$	$6/8$	$9/12$	$12/16$

OR: Divide both numerator and denominator by the SAME number

Original	Divide by 2	Divide by 3	Divide by 4
$4/6$	$2/3$	-	-
$6/9$	-	$2/3$	-
$8/12$	$4/6$	-	$2/3$
$12/16$	$6/8$	$4/4$	$3/4$

6. Reciprocals

Definition: Two numbers whose product is 1 **How to find:** Flip the numerator and denominator

Number	Reciprocal	Check (Product = 1)
$\frac{2}{3}$	$\frac{3}{2}$	$\frac{2}{3} \times \frac{3}{2} = \frac{6}{6} = 1$
$\frac{3}{4}$	$\frac{4}{3}$	$\frac{3}{4} \times \frac{4}{3} = \frac{12}{12} = 1$
$\frac{1}{5}$	$\frac{5}{1}$ (or 5)	$\frac{1}{5} \times 5 = 1$
2	$\frac{1}{2}$	$2 \times \frac{1}{2} = 1$
5	$\frac{1}{5}$	$5 \times \frac{1}{5} = 1$
$1\frac{1}{2}$ ($\frac{3}{2}$)	$\frac{2}{3}$	$\frac{3}{2} \times \frac{2}{3} = \frac{6}{6} = 1$
$2\frac{3}{4}$ ($\frac{11}{4}$)	$\frac{4}{11}$	$\frac{11}{4} \times \frac{4}{11} = \frac{44}{44} = 1$

Special Cases:

- Reciprocal of a whole number: Write it as a fraction then flip — $3 = \frac{3}{1} \rightarrow \text{reciprocal} = \frac{1}{3}$; $7 = \frac{7}{1} \rightarrow \text{reciprocal} = \frac{1}{7}$
- Reciprocal of a mixed number: First convert to improper fraction, then flip — $2\frac{1}{2} = \frac{5}{2} \rightarrow \text{reciprocal} = \frac{2}{5}$

7. Addition and Subtraction — Basic Rules

Rule 1: If denominators are EQUAL — Add or subtract the numerators, keep the denominator the same

Type	Example	Answer
Addition	$\frac{1}{5} + \frac{4}{5}$	$\frac{1}{5} + \frac{4}{5} = \frac{(1+4)}{5} = \frac{5}{5} = 1$
Addition	$\frac{2}{7} + \frac{3}{7}$	$\frac{(2+3)}{7} = \frac{5}{7}$
Subtraction	$\frac{5}{6} - \frac{2}{6}$	$\frac{(5-2)}{6} = \frac{3}{6} = \frac{1}{2}$
Subtraction	$\frac{7}{12} - \frac{3}{12}$	$\frac{(7-3)}{12} = \frac{4}{12} = \frac{1}{3}$

Rule 2: If denominators are UNEQUAL — Convert to equivalent fractions with the same denominator (LCD/LCM), then add or subtract numerators

8. Multiplication of Fractions

Rule: Multiply numerators together, multiply denominators together

Example	Calculation	Answer
$\frac{1}{2} \times \frac{1}{3}$	$(1 \times 1) / (2 \times 3)$	$\frac{1}{6}$
$\frac{2}{5} \times \frac{1}{3}$	$(2 \times 1) / (5 \times 3)$	$\frac{2}{15}$
$\frac{3}{4} \times \frac{2}{5}$	$(3 \times 2) / (4 \times 5) = \frac{6}{20}$	$\frac{3}{10}$
$\frac{2}{3} \times \frac{5}{8}$	$(2 \times 5) / (3 \times 8) = \frac{10}{24}$	$\frac{5}{12}$

With Mixed Numbers: Convert mixed numbers to improper fractions first, then multiply

Mixed Number Multiplication	Calculation	Answer
$1 \frac{1}{2} \times 1 \frac{3}{4}$	$\frac{3}{2} \times \frac{7}{4} = \frac{21}{8}$	$2 \frac{5}{8}$
$2 \frac{1}{3} \times 1 \frac{1}{2}$	$\frac{7}{3} \times \frac{3}{2} = \frac{21}{6}$	$3 \frac{1}{2}$
$3 \frac{2}{5} \times 2 \frac{1}{4}$	$\frac{17}{5} \times \frac{9}{4} = \frac{153}{20}$	$7 \frac{13}{20}$

Cancellation Method: Cancel common factors before multiplying — makes numbers smaller and easier. Example: $\frac{3}{8} \times \frac{4}{9} \rightarrow$ cancel 3 with 9 (divide both by 3): $\frac{1}{8} \times \frac{4}{3} \rightarrow$ cancel 4 with 8 (divide both by 4): $\frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$

9. Division of Fractions

Rule: Multiply by the reciprocal of the second fraction

Division Problem	Convert to Multiplication	Answer
$\frac{2}{3} \div \frac{3}{4}$	$\frac{2}{3} \times \frac{4}{3}$	$\frac{8}{9}$
$\frac{1}{5} \div \frac{2}{5}$	$\frac{1}{5} \times \frac{5}{2}$	$\frac{1}{2}$
$\frac{3}{4} \div \frac{1}{2}$	$\frac{3}{4} \times \frac{2}{1}$	$\frac{3}{2} = 1 \frac{1}{2}$
$\frac{6}{7} \div 3$	$\frac{6}{7} \times \frac{1}{3}$	$\frac{2}{7}$
$8 \div \frac{4}{5}$	$8 \times \frac{5}{4}$	10

Quick Trick: “Keep, Change, Flip” — KEEP the first fraction · CHANGE division to multiplication · FLIP the second fraction (find reciprocal)

10. The “Of” Operation

Key Rule: “OF” means “MULTIPLY” ($\times$)

Expression	Meaning	Calculation
$\frac{1}{2}$ of 100	$\frac{1}{2} \times 100$	50
$\frac{1}{3}$ of 20 kg	$\frac{1}{3} \times 20 \text{ kg}$	$\frac{20}{3} = 6 \frac{2}{3} \text{ kg}$
$\frac{1}{2}$ of $\frac{1}{3}$	$\frac{1}{2} \times \frac{1}{3}$	$\frac{1}{6}$
$\frac{1}{3}$ of $\frac{4}{5}$	$\frac{1}{3} \times \frac{4}{5}$	$\frac{4}{15}$
$\frac{2}{3}$ of $1 \frac{3}{4}$	$\frac{2}{3} \times \frac{7}{4}$	$\frac{14}{12} = \frac{7}{6} = 1 \frac{1}{6}$

Important: In BODMAS, “OF” has a higher priority than division and multiplication!

11. BODMAS Rule — Order of Operations

Letter	Stands For	Priority
B	Brackets	1st (Highest)
O	Of (or Orders/Powers)	2nd
D	Division	3rd
M	Multiplication	3rd (Same as D, left to right)
A	Addition	4th
S	Subtraction	4th (Same as A, left to right)

BODMAS Order: $B \rightarrow O \rightarrow D/M \text{ (left to right)} \rightarrow A/S \text{ (left to right)}$

B - Brackets
O - Of
D - Division
M - Multiplication
A - Addition
S - Subtraction

“Please Excuse My Dear Aunt Sally” (Common English mnemonic) — P = Parentheses (Brackets), E = Exponents (Orders), M = Multiplication, D = Division, A = Addition, S = Subtraction

12. Key Symbols

Symbol	Meaning	Example
+	Addition	$1/5 + 3/5 = 4/5$
–	Subtraction	$3/4 - 1/4 = 1/2$
×	Multiplication	$1/2 \times 1/3 = 1/6$
÷	Division	$3/4 \div 1/2 = 1\ 1/2$
of	Multiplication	$1/2$ of 100 = 50
=	Equals	$1/2 + 1/2 = 1$
()	Brackets	$(1/2 + 1/3) = 5/6$
{ }	Curly Brackets	$2\{3 + 1\} = 8$
[]	Square Brackets	$[2 + 3] = 5$

13. Quick Reference Table

Type	Example	Rule
Equivalent fractions	$1/2 = 2/4 = 3/6$	Multiply/divide both parts by same number
Mixed → Improper	$1\ 1/2 = 3/2$	$(W \times D + N)/D$
Improper → Mixed	$7/3 = 2\ 1/3$	Divide $N \div D$, remainder becomes numerator
Add/Subtract (same denominator)	$1/5 + 2/5 = 3/5$	Add/subtract numerators, keep denominator
Add/Subtract (different denominators)	$1/2 + 1/3 = 3/6 + 2/6 = 5/6$	Find LCM, convert, then add/subtract
Multiply	$1/2 \times 3/4 = 3/8$	Multiply numerators, multiply denominators
Divide	$1/2 \div 3/4 = 1/2 \times 4/3 = 2/3$	Keep, Change, Flip
Reciprocal	Reciprocal of $3/4 = 4/3$	Flip numerator and denominator
"of"	$1/2$ of 100 = 50	Replace "of" with ×

14. Check Yourself — Quick Questions

No.	Question	Answer
1	What is the numerator in $\frac{3}{4}$?	3
2	What is the denominator in $\frac{3}{4}$?	4
3	Is $\frac{2}{3}$ a proper or improper fraction?	Proper ($2 < 3$)
4	Is $\frac{5}{3}$ a proper or improper fraction?	Improper ($5 > 3$)
5	What is the reciprocal of $\frac{2}{3}$?	$\frac{3}{2}$
6	Convert $2\frac{3}{4}$ to an improper fraction	$\frac{11}{4}$
7	Convert $\frac{7}{2}$ to a mixed number	$3\frac{1}{2}$
8	What is $\frac{1}{2} + \frac{1}{2}$?	1
9	What is $\frac{1}{2} \times \frac{1}{3}$?	$\frac{1}{6}$
10	What is $\frac{1}{2}$ of $\frac{1}{3}$?	$\frac{1}{6}$
11	What does "O" stand for in BODMAS?	Of
12	What is the reciprocal of 5?	$\frac{1}{5}$
13	What is $\frac{1}{2} \div \frac{1}{3}$?	$1\frac{1}{2}$
14	Are $\frac{1}{2}$ and $\frac{2}{4}$ equivalent?	Yes
15	What is $\frac{2}{3} - \frac{1}{3}$?	$\frac{1}{3}$

15. Quick Memory Aids

Fraction Vocabulary

- Numerator → Number on TOP (“N” for “North”)
- Denominator → Number on BOTTOM (“D” for “Down”)

Mixed to Improper

“Multiply, Add, Keep” → $(W \times D + N) / D$

Improper to Mixed

“Divide, Remainder, Keep” → Quotient + (Remainder/Divisor)

Multiply Fractions

“Top × Top, Bottom × Bottom”

Divide Fractions

“Keep, Change, Flip” → $3/4 \div 1/2 = 3/4 \times 2/1$

BODMAS

“Brackets, Of, Division, Multiplication, Addition, Subtraction”

“of” = ×

Always replace “of” with multiplication!

To Find Reciprocal

“Turn it upside down!”

Revision Tip: Practice converting between mixed numbers and improper fractions daily. Master these basics before moving to addition, subtraction, multiplication, and division!